

# **Analyzing the Role of AI-Assisted Programming in Modern Software Development**

**S17535**

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## **Task 1: Conceptual Understanding (Report)**

### **What is AI-Assisted Programming?**

AI-Assisted Programming means using Artificial Intelligence to help write, understand, fix, and improve computer code. AI can write code, fix errors (debugging), explaining code, finding bugs, creating functions, writing comments, generate documentation and suggesting improvements

### **Types of AI coding tools (e.g., code generation, debugging, refactoring)**

1. Code Generation - AI writes code based on our instructions.
2. Debugging - Debugging means finding and fixing errors (bugs) in our code.
3. Refactoring - Refactoring means improving the structure of our code without changing what it does.
4. Code Completion - AI predicts what to type and completes it.
5. Code Explanation - AI explains what a piece of code does in simple language.
6. Documentation Generation- Documentation explains how code works. AI can create comments and documentation automatically.
7. Code Translation - Documentation explains how code works. AI can create comments and documentation automatically.
8. Testing Assistance - Documentation explains how code works. AI can create comments and documentation automatically.

### **Advantages in software development**

- Faster Code Development - AI can generate code in seconds, reducing the time needed to write programs.
- Faster debugging - AI detects syntax errors, logical mistakes, and common programming bugs.
- Improved code quality - AI suggests cleaner, more organized, and easier-to-read code.
- Easier maintenance —updating, fixing, or improving software after it has been developed.
- Generates Documentation - AI creates comments and documentation for code automatically.

## Risks and limitations (e.g., hallucinated code, dependency, security issues)

- Hallucinated Code - Sometimes AI generates incorrect, fake, or non-working code while sounding very confident. This is called hallucination.
- Security Issues - AI may generate code that contains security weaknesses or suggest unsafe coding practices.
- Dependency on AI - Some programmers rely on AI for everything instead of learning programming themselves.
- Limited AI Credits - Many AI coding tools provide a **limited number of uses (credits)**. Each time you ask AI to generate code, explain something, or perform a task, it may consume some credits.

## Task 2: Practical Implementation (Include screenshots to the Report)

### Library Management System

A Library Management System is a software application used to manage books, members, and borrow activities in a library

Version A - LibraryManagementSystem.java

Some code basics take from - <https://www.w3schools.com/java/default.asp>

Version B – AI.java

# AI prompt - I need a java code for Library Management System. It does add books, view books, add student, view student, return books and barrow books give minimal code 1 file use oops Concept also.

GitHub- [https://github.com/vithusan1225/IT3003\\_A2\\_AI-Assisted-Programming.git](https://github.com/vithusan1225/IT3003_A2_AI-Assisted-Programming.git)

# Both code in above GitHub repository

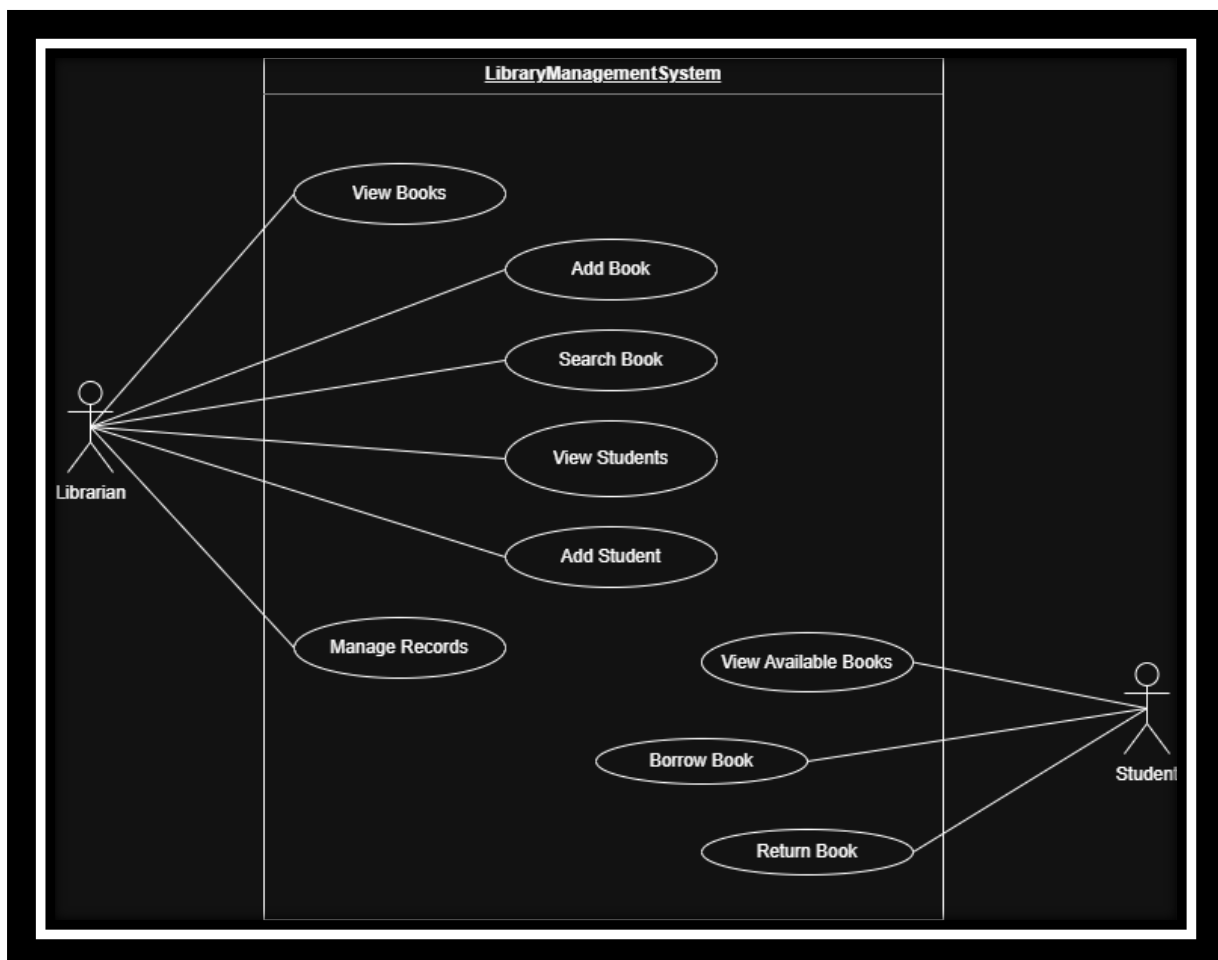
## Compare both implementations

| Criteria(properties) | My code                                                   | AI-Assisted Programming                                 |
|----------------------|-----------------------------------------------------------|---------------------------------------------------------|
| Development Time     | Tack long time to develop                                 | Faster because AI generates code templates              |
| Code Quality         | Depends on my experience                                  | Usually structured but requires checking                |
| Error Handling       | I need to identify and fixes errors manually              | AI can suggest exception handling methods               |
| Readability          | Depends on my coding style (only I can read easily)       | AI can generate cleaner formatting with proper comments |
| Maintainability      | Person needs to know about good software design knowledge | AI can suggest refactoring techniques                   |

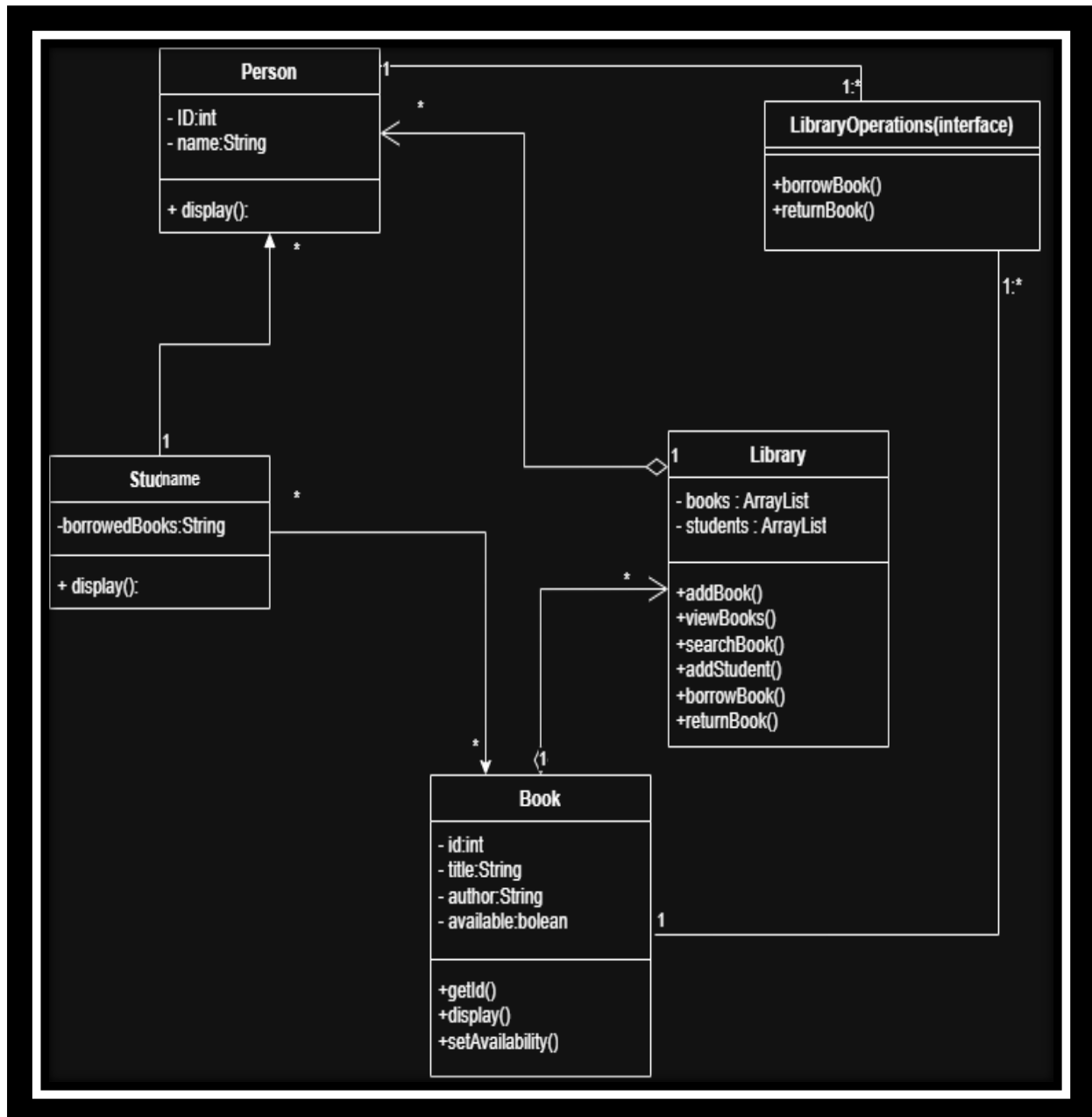
## Diagrams

# All diagram also in above GitHub repository

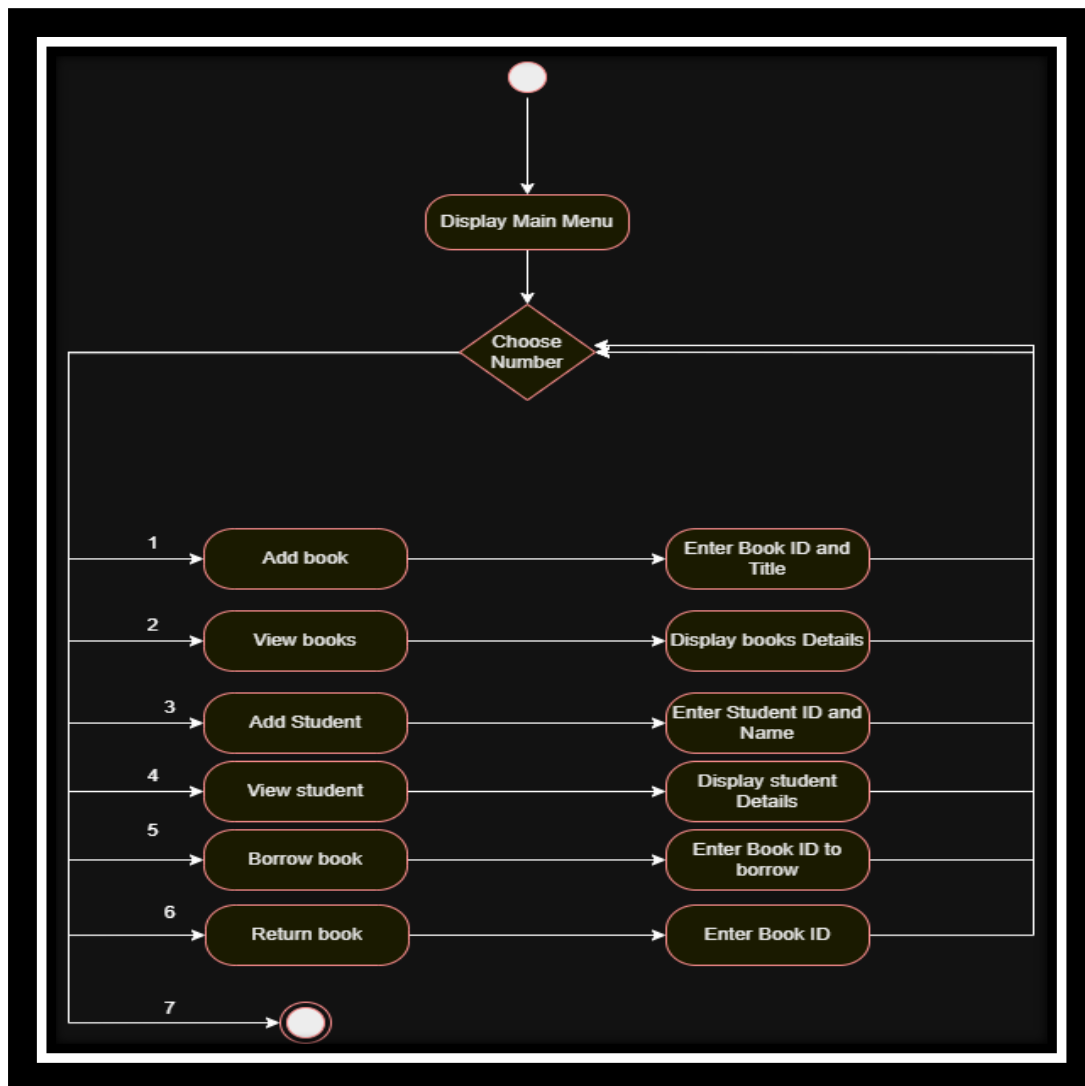
### Use Case Diagram



## Class Diagram



## Activity Diagram



| Feature            | Version A (Manual Code) | Version B (AI-Assisted Code)          |
|--------------------|-------------------------|---------------------------------------|
| Book management    | Manually created        | Improved with encapsulation           |
| Student management | Manually created        | Improved using inheritance            |
| Borrow/Return      | Basic implementation    | Student-book relationship added       |
| Search             | Not available           | Added using AI suggestion             |
| Date tracking      | Not available           | Added using Local Date                |
| Error handling     | Limited                 | Added try-catch                       |
| Data security      | Public variables        | Private variables + getters/setters   |
| OOP concepts       | Basic classes           | Interface, inheritance, encapsulation |
| Code quality       | Simple                  | More structured and maintainable      |

# This table Initial idea takes from AI.

AI prompt- I upload my code (LibraryManagementSystem.java) and AI generated (AI.java) code. Read that and perform a compare Which code sections are different among that.

## **Task 3: Critical Analysis of AI-Assisted Programming**

### **How AI changed my coding approach**

Before using AI, I usually started coding by thinking about the problem and writing the program step by step. With AI, my coding approach changed because I can get ideas, examples, and explanations faster. AI helps me understand how to structure my code, find possible solutions, and fix errors quickly.

Spending a lot of time searching for solutions, I can ask AI for guidance and then modify the code according to my requirements. This helped me focus more on understanding the logic and design of the program.

### **Whether AI improved or reduced my understanding**

AI improved my understanding when I used it as a learning tool. For example, AI can explain difficult programming concepts, suggest better coding methods, and explain errors in a simple. It helped me learn OOP concepts, debugging techniques, and program structure.

However, if I only copy and paste AI-generated code without studying it, my understanding can decrease. I may have worked code but not know how it works. Therefore, AI is useful only when I read, test, and understand the generated code.

### **Cases where AI should NOT be used**

AI should not be used for everything. It should not replace personal learning and problem-solving skills. Students should avoid using AI to complete entire assignments without understanding because it can reduce their programming ability. (my important point) Use Grammarly AI tool for good format of writing}

AI-generated code should also not be used directly in important systems without testing because AI may produce incorrect, inefficient, or insecure code. Developers should always check and improve AI suggestions. (Reduce Security)

### **Ethical considerations in AI-generated code**

(I can't understand this question trying to write good ideas)

There are some important things to think about when using AI-generated code. We should not just copy AI code and say it is fully our own work because we may not understand how it works. If we use AI, we should be honest about it when needed.

Sometimes AI can create code from existing examples, so there can be problems about who owns that code. Also, AI code can have mistaken or security issues. Therefore, programmers should always check, test, and understand the code before using it in a real project.

## Conclusion

AI is a powerful tool that can improve programming speed and learning. However, it should be used as an assistant, **not as a replacement for human thinking**. The best approach is to use AI for guidance, understand the generated code, and develop independent programming skills.

## Task 4: Conclusion

In my opinion, AI is a very useful tool for programming education and the software industry. It helps students learn faster by explaining concepts, finding errors, and providing examples. For developers, AI can save time and improve productivity by helping with coding and problem-solving.

However, AI should not replace human thinking and learning. Students and programmers should understand the code instead of blindly copying AI-generated solutions. AI should be used as a helper to improve skills, not as a replacement for programming knowledge.

Overall, AI has a positive impact on programming when it is used responsibly. The best way is to combine human creativity and problem-solving skills with the support provided by AI.

GitHub Link - [https://github.com/vithusan1225/IT3003\\_A2\\_AI-Assisted-Programming.git](https://github.com/vithusan1225/IT3003_A2_AI-Assisted-Programming.git)

#This repository contain my and AI cods , diagram Drawio files and report word file.